COMP 018: Database Administration - Activity 6 Progress Report
Database Backup Planning, Verification, Restoration, and Disaster Recovery Simulation

Project Identification & Metadata

Course : COMP 018 - Database Administration
Evaluating Group : Group 4: Olivarez, Abangan, Pocsidio, Infantado, and Bartolata
Evaluating Group Repository : https://github.com/Nycri/brewpos
Target Project Monitored : OJT Management System (Project-OJT-RMA-System)
Target Project Group : Group 3 (Evaluated under Tracker Key: Group 06 / Group 03): Simon, De Dios, Ramos, Dela Rosa, Sietereales, Geraldez
Target Project Repository : https://github.com/swvenz/Project-OJT-RMA-System (Branches: main, ojt-system-untested.demo)
Primary Database : ojt_management (PostgreSQL 18.6 on x86_64 Windows, compiled by MSVC)
Restored Test Database : ojt_management_restore_test

ACTIVITY 6: Database Backup, Restoration, and Recovery

3.1 Backup Planning & Architecture

The disaster recovery plan for the OJT Management System database encompasses twelve relational tables within the public schema, capturing user profiles, audit logs, student rosters, partner company records, supervisor accounts, compliance requirements, applications, placements, attendance logs, and evaluation grades. The plan covers sequence counters, primary keys, foreign keys, constraints, and privilege grants to ensure operational continuity.

The backup strategy utilizes PostgreSQL custom compressed archive format via a dedicated backup operator role. Archive dumps are stored in a protected directory outside web roots and repository trees. Recovery objectives define a Recovery Point Objective of 1 hour and a Recovery Time Objective of 30 minutes. Retention guidelines specify maintaining seven daily backups alongside four weekly snapshots.

Because database login accounts exist at the PostgreSQL cluster level and are not stored within single-database dumps, disaster recovery plans require executing access-control role scripts on new server instances prior to restoring database contents.

3.2 Backup Creation, Checksum & Archive Inspection

Logical backup creation was executed using the dedicated backup account, producing a compressed custom format archive sized at 56,714 bytes. Cryptographic verification computed a SHA-256 hash value of FC979FF8640AE61F9A747475427E472B7D8344B193D751097DDEF576E3A10A7F,

establishing a baseline for integrity verification.

Inspection of the archive table of contents verified 143 catalog entries, confirming the inclusion of all twelve database tables, sequence objects, primary keys, check constraints, foreign keys, and default privilege grants. To ensure recoverability beyond file inspection checks, a complete restoration drill was conducted in an isolated test database.

3.3 Restoration into Separate Test Database
An isolated test database named ojt_management_restore_test was created on the local server instance. The custom archive dump file was restored into this database using administrative utility tools. The restoration process completed cleanly with zero error codes.

3.4 Post-Restore Verification Suite
Post-restoration verification confirmed that all twelve relational tables were recreated with proper schema structure under administrative ownership. Record count validation across tables demonstrated complete parity with the source database, matching two academic supervisors, four attendance entries, two audit logs, three companies, one company supervisor, one evaluation, two applications, one placement, five requirements, four student requirements, two students, and seven user profiles. Database checks confirmed that all 129 constraints were fully restored.

Access control verification on the restored test database confirmed that privilege definitions transferred accurately. Queries executed under the application runtime account successfully returned company records, while query attempts by the read-only auditing account on user credential tables were blocked by the database security layer.

3.5 Controlled Recovery Simulation
A controlled data recovery simulation was performed to test incident response procedures. A disposable requirement record was inserted into the restored test database to establish a baseline state. A data loss event was then simulated by removing the record, and queries confirmed its deletion.

Targeted record restoration was executed to reinsert the missing data row without affecting existing database entries. Post-recovery verification confirmed that the record was fully restored with its original attributes intact. Following verification, the test record was removed to clean up the test environment.

3.6 Limitations, Evaluation & Production Recommendations
Restoration procedures on fresh server installations require pre-executing role definition scripts to establish user accounts before restoring database ACLs. Because student requirements reference file upload paths, database backups must be paired with filesystem backups of upload directories. For production operations, daily automated backup generation should be scheduled using system task schedulers to archive database dumps with date-stamped filenames and

retain snapshots according to policy. Enabling continuous write-ahead log archiving will support point-in-time recovery capabilities.

CLASS GOOGLE SHEETS TRACKER ENTRY (Activity 6)

Column Header	**Value**
Project	Group 3 - OJT Management System (Project-OJT-RMA-System) [Tracker Key: Group 06 / Group 03]
Database and backup file	ojt_management, ojt_management_activity6.dump, Size: 56,714 bytes, SHA256: FC979FF8640AE61F9A747475427E472B7D8344B193D751097DDEF576E3A10A7F
Backup method	pg_dump custom format (-Fc), compressed binary dump (2026-11-05 08:07:30)
Restore test result	Success (Restored into ojt_management_restore_test; all 12 tables, 129 constraints, and data verified)
Access control after restore	(Activity 4 roles and permissions validated; ojt_report_user denied reading users)
Recovery simulation result	Recovered (Simulated deleted test requirement successfully restored from verified baseline state)
Proposed RPO and RTO	RPO: 1 hour, RTO: 30 minutes
Evidence folder link	https://github.com/Nycri/brewpos/tree/main/activity-6/Evidence-act6
Feedback sent to owners	(05/11/2026)
PDF progress document	Activity6_Progress_Report_Group3.pdf
Last updated	2026-11-05

AI USAGE DECLARATION

Google Gemini 3.8 Flash was utilized during project evaluation to inspect database schema definitions, evaluate relational hierarchy structures, formulate role verification test scenarios, analyze database maintenance operations and query performance execution plans, verify backup creation and restoration procedures, and structure technical report documentation. All database statements, configuration checks, and verification procedures were directly executed and validated against PostgreSQL 18.6 within the local database environment.